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Week 01: Introduction to Cyber Security and Ethical Hacking

01: The CIA Triad: CIA stands for **confidentiality**, **Integrity** and **Availability**. Complete cyber security is based on 3 principles these are developed in 1980 so that every non technical person also understands the cyber security in a simplified way.

1) **Confidentiality:** This ensures that information is accessed only by authorized people.

Example: No one should be able to access some one's personal mobile phone

2) **Integrity:** Data remains same, accurate unaltered from sending end to receiving end.

Example: User A transfer email to user B should receive email that user A sends without any changing

3) **Availability:** Ensures that system and information remains available for authorized users when needed.

Example: Websites, apps, and servers should be up and running 24/7.

02: Types of Hackers

There are mainly 3 types of hackers **Black Hat**, **White Hat** and **grey hat hackers**.

- **Black Hat Hacker:** A type of hacker that breaks someone's system, application, network without permission for malicious purposes **E.g** to harm someone's reputation, blackmail, financial damage, data exfiltration etc.
- **White Hat Hacker:** A type of hacker that hacks system network application within permission to test and fix all the vulnerabilities to secure the digital infrastructure before malicious actors reach and cause harm.
- **Grey Hat Hacker:** Skilled hackers operating between ethical and unethical boundaries. They do hacking for recognition and curiosity mostly. Some times they report the vulnerabilities and sometimes not Usually they don't have any malicious intents.

03: Phases of Ethical Hacking

1. **Reconnaissance:** Reconnaissance is the initial phase of ethical hacking where attackers gathers information as much as they can about the target system, network or application. It is further divided in **active reconnaissance** and **passive reconnaissance**. In **active reconnaissance** attacker gets directly involved with target such as sending network probes or making direct web request to check system is alive or not whereas in **passive reconnaissance** attackers gets information from **OSINT** without interacting directly with target.
2. **Scanning:** Scanning involves active devices, open ports and services on the target network. Scanning helps attacker to find potential vulnerabilities in a target system or network. Vulnerabilities can be scanned via automated tools or manually.

3. **Gaining Access:** Hackers exploit the possible vulnerabilities that they discovered of a target system, network or application to gain unauthorized access.
4. **Maintaining Access:** Once access is gained, the hacker must maintain a foothold in a target system to perform further actions such as escalating privileges, lateral movement data exfiltration and so on. For maintaining access they often use backdoors and rootkits to maintain their ownership in a target system.
5. **Clearing Tracks:** To remain undetectable as a hacker gets initial access to a system he must disable the logs so that all malicious activities that he will be performing will not get detected and the hacker remains safe.

04: Penetration Testing vs Vulnerability Assessment

Vulnerability Assessment: Vulnerability assessment means weakness assessment. It is a process of identifying and reporting all the vulnerabilities or weakness in a system, network or application. It does not exploit them. Only tells that this is the weakness that needs to be fixed as soon as possible.

Penetration Testing: Penetration testing is the process of legally testing a computer system, network or application by exploiting vulnerabilities. The purpose is to show the stakeholders the real impact of each vulnerability, what disaster could happen if not fixed, and an attacker could exploit this and can be used for bad intentions.

05: Try Hack me rooms

Pentesting Fundamentals

Share your win with your peers

Your progress can inspire others in your community.



Pentesting Fundamentals completed!

asadfarzand711 completed another room on their cyber security journey.

Completed tasks	Points earned	Streak
📋 5	🏆 72	🔥 1

Share on social media

Share your win before starting the next room

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Introduction to Offensive Security

Understand what Offensive Security entails, and practice breaking into computer systems by exploiting applications and networks.

Start by learning the basics of web application security and practice exploiting a vulnerability in an online website. Investigate operating system security and break into a target computer. Finally, you will explore network security and hack into an online server.



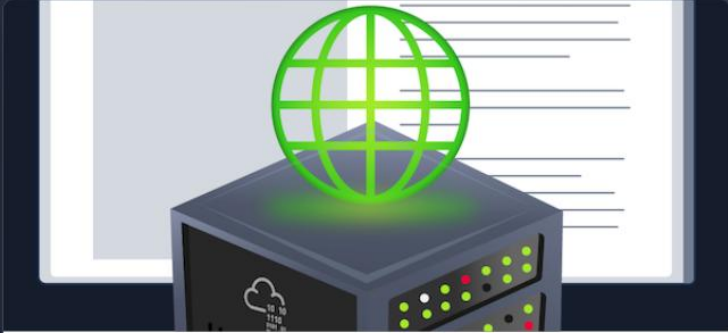
**Web Application Security** 
Learn about web applications and explore some of their common security issues.

**Operating System Security** 
This room introduces users to operating system security and demonstrates SSH authentication on Linux.

**Network Security** 

Need to know
**Introduction to Cyber Security**
Understand what is offensive and defensive security, and learn about careers available in cyber.





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06 Conclusion: This task is based on the fundamentals of ethical hacking the terminologies used such as CIA triad, difference between vulnerability assessment and penetration testing types of hackers phases of ethical hacking. This task helps to strengthen the foundation for further task